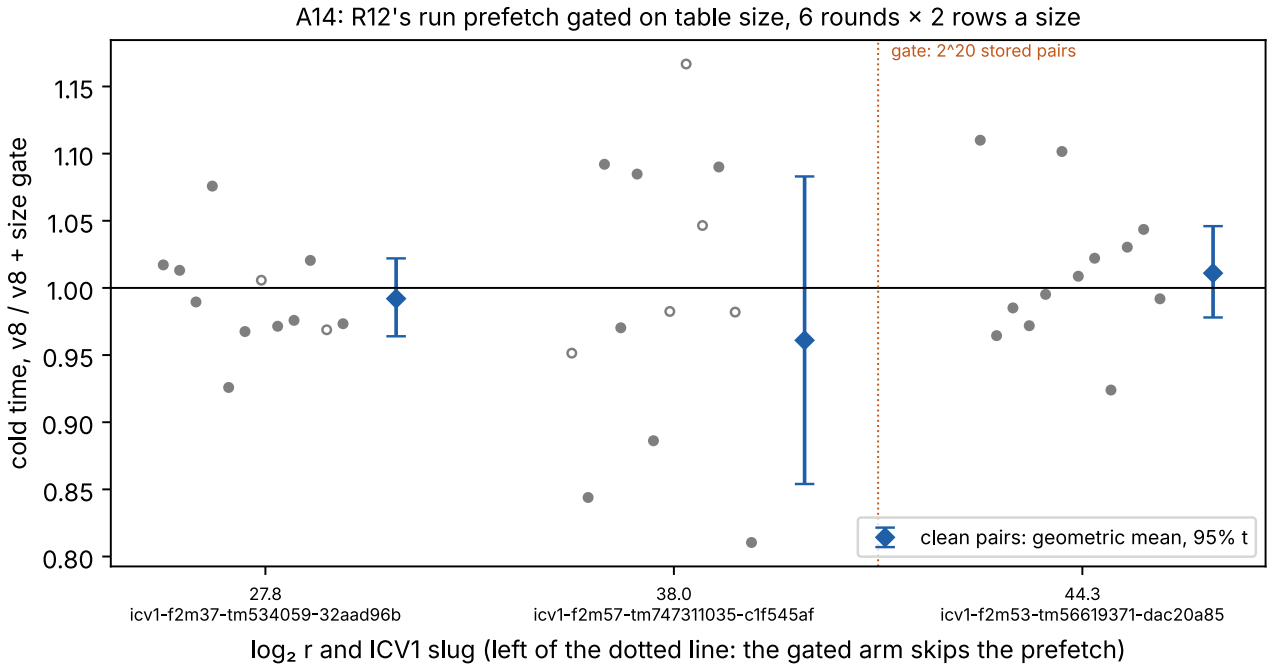
A14: R12′s run prefetch gated on the table's size

Exploration, no round declared. v8 (`ic-r12-v7-23138d20`, R12′s candidate, now the baseline) against v8 with the prefetch pass skipped below 2^20 stored pairs (`ic-r13-v8-ebfd2d2f`, `size-gate.patch`). Run 2026-10-17 03:10–03:28 UTC on the Emerald Rapids host R12 ran on: 72 isolated processes on CPU 2, six rounds with the order alternating, `M1`′s two rows at three sizes. Records: `research/ic_tool_program/explorations/A14-prefetch-size-gate-20261017/` (`pairs.tsv`, `summary.tsv`, `phases.tsv`, `runs.tar.xz`, `explore.log`).

- **Completed:** 72 of 72. Seven v8 runs were contended and are left out of the intervals. The busy gate refused 201 starts before any run began, and every refused run was asked for again.
- **Logarithms:** every pair recovered the same logarithm in both arms (36 of 36).
- **Outcome:** below the gate the gated arm is no faster: 0.992 at 2^27.8 and 0.961 at 2^38.0, both intervals including 1. Above the gate it runs v8′s code (1.011). The gate cannot reach the 1.03 line at the target sizes, so no round is declared and v8 stays the baseline.



research/ic_tool_program/explorations/A14-prefetch-size-gate-20261017/pairs.tsv; 72 isolated processes; open circles: a pair with a contended run

Figure 1: Cold time, v8 over the gated arm, per pair (open circles: a contended run) with the clean pairs' geometric mean and 95% t interval. Values from `pairs.tsv` and `summary.tsv`.

Cold time, v8 / gate

curve	$\log_2$ r	prefetch in gate	clean pairs	cold	95%	all pairs
`icv1-f2m37-tm534059-32aad96b`	27.8	skipped	10	0.992	[0.964, 1.022]	0.991
`icv1-f2m57-tm747311035-c1f545af`	38.0	skipped	7	0.961	[0.854, 1.083]	0.987
`icv1-f2m53-tm56619371-dac20a85`	44.3	runs	12	1.011	[0.978, 1.046]	1.011

Phases, v8 / gate (geometric mean of 12 pairs' median repetitions)

curve	collection	build	selection
`icv1-f2m37-tm534059-32aad96b`	1.003	0.975	0.993
`icv1-f2m57-tm747311035-c1f545af`	1.029	0.998	1.001
`icv1-f2m53-tm56619371-dac20a85`	1.010	1.027	0.993

Median cold time over 12 processes, v8 → gate: 3.09 → 3.09 ms, 38.97 → 38.65 ms and 406.37 → 398.84 ms.

Scope

One host, one core, three sizes, two rows a size. At 2^38.0, skipping the pass makes collection 2.9% faster, about 1% of cold time. This exploration cannot resolve that. R12′s reading of 0.971 [0.950, 0.992] at 2^27.8 is not recovered (0.992 [0.964, 1.022]). That does not show R12′s small-size readings were noise. Its own A/A rule already placed them inside its band.